



Seat No. \_\_\_\_\_

No. of Printed Pages: 02

[26]

**SARDAR PATEL UNIVERSITY**  
**External Examination – October 2025**  
**BCA (Semester-V) – (NEP 2020)**  
**[US05MABCA02] : Visual Programming**

**Date: 06/10/2025 (Monday)**

**Time: 10.00am to 12.30pm**

**Total Marks: 50**

**Note: Figure indicates at right is maximum marks for question.**

**Q -1 Select an appropriate option for the following statements. [10]**

1. Which of the following is NOT a .NET language?  
[a] VB.NET [b] C#  
[c] Python [d] F#
2. What is the .NET Framework Class Library?  
[a] A set of hardware APIs [b] A collection of reusable classes and interfaces  
[c] A .NET plugin [d] A database system
3. Which operator is used for concatenating strings in VB.NET?  
[a] & [b] +  
[c] % [d] &&
4. Which loop is best when the number of iterations is known?  
[a] Do..loop [b] While  
[c] For [d] With
5. Which keyword is used to call a procedure?  
[a] Call [b] Use  
[c] Invoke [d] Run
6. Which control is used to accept text input from the user?  
[a] Label [b] Button  
[c] TextBox [d] GroupBox
7. Which control is used to allow users to select a date?  
[a] TimePicker [b] Calander  
[c] DateTimePicker [d] Timer
8. Which dialog box is used to choose fonts and styles?  
[a] FontDialog [b] StyleDialog  
[c] FontChooser [d] RichTextEditor
9. In disconnected architecture, data is stored in:  
[a] SqlDataReader [b] DataStream  
[c] DataSet [d] SqlCommand
10. Which object represents a connection to a SQL Server database?  
[a] SqlCommand [b] SqlAdapter  
[c] SqlConnection [d] SqlReader



**Q:2** Explain the major components of the .NET architecture. [10]

**OR**

**Q:2** Explain the different types of operators available in VB.NET with examples of each category. [10]

**Q:3** Compare and contrast SDI (Single Document Interface) and MDI (Multiple Document Interface) applications. Explain how to create and manage MDI parent and child forms in VB.NET. [10]

**OR**

**Q:3** Explain the syntax and usage of the If...Then...Else statement in VB.NET with an example. How is Select Case different from multiple If...ElseIf statements? Give a suitable example. [10]

**Q:4** What is the purpose of advanced controls like LinkLabel, RichTextBox, and TreeView in desktop applications? [10]

**OR**

**Q:4** Explain the concept of exception handling in Windows Forms applications with specific reference to structured exception handling using try...catch...finally. [10]

**Q:5** Define ADO.NET. Also explain the connected architecture of ADO.NET. [10]

**OR**

**Q:5** What are the advantages of using reports in VB.NET applications? Write a step-by-step process to design a printable report using Microsoft Report Viewer in VB.NET. [10]

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